

Chair of Computer Architecture (M. Sc.)

Model Enumeration using IPASIR-UP

Master's Thesis

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Working Period

start date - end date

Abstract

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1 Introduction

The Boolean satisfiability problem (SAT) is defined as follows:

Given a propositional formula F over variables $V = \{v_1, \dots, v_n\}$, decide whether there exists an assignment $\alpha : V \rightarrow \{0, 1\}$ such that $\alpha \models F$.

This problem was the first to be proven NP-complete by Cook [1] in 1971 and independently by Levin [2] in 1973. As a result, finding efficient algorithms for the SAT problem quickly became one of the most important and influential challenges in computer science. Modern Conflict-Driven Clause Learning (CDCL) solvers are capable of handling formulas with millions of variables and clauses. As a result, SAT solving plays a central role in hardware and software verification, automated reasoning, planning, scheduling, model checking, and cryptanalysis.

State-of-the-art solvers such as CaDiCal, Glucose, and Kissat demonstrate how theoretical advances combined with low level optimizations can transform an intractable worst-case problem into a powerful practical tool. These solvers typically focus on the decision problem: determining whether at least one satisfying assignment exists. If the formula is satisfiable, the solver returns a single model.

However, many applications require more than just one solution. In areas such as configuration, diagnosis, quantitative reasoning, explanation generation, and formal verification, it is often necessary to enumerate all satisfying assignments or a specific subset of them. This task, known as model enumeration (or All-SAT), generalizes the classical SAT problem from computing a single satisfying assignment to computing the whole set $\{\alpha : V \rightarrow \{0, 1\} \mid \alpha \models F\}$. While conceptually straightforward, model enumeration introduces significant algorithmic challenges. The number of satisfying assignments may be exponential in the number of variables, and naive approaches can quickly become infeasible.

A common technique for model enumeration is the iterative use of a SAT solver combined with blocking clauses. After each model is found, a clause composed of the negation of every literal in that model is added to the CNF formula, ensuring that the solver does not rediscover the same assignment. With the introduction of standardized interfaces such as IPASIR, SAT solvers can be seamlessly integrated into larger systems without the need to maintain the solver itself. More recently, IPASIR-UP was introduced, expanding IPASIR with User Propagators. This allows adding clauses without restarting the search. Using IPASIR-UP with Blocking Clauses yields a very clean model enumerator, that benefits from the underlying solver CaDiCal while CaDiCal itself remains unchanged. Although effective, this approach has several drawbacks. The repeated addition of blocking clauses increases the size of the formula, thereby degrading solver performance by affecting propagation efficiency and memory consumption. As the number of models grows, these effects scale up, eventually making blocking clauses infeasible.

In 2024 [3] presented TabularAllSat: A Sat solver that is especially designed for

disjoint projected enumeration for SAT and SMT without blocking clauses. TabularAllSat outperforms all other state-of-the-art AllSAT solvers. This was achieved by "combining the best of current All- SAT state-of-the-art literature": (i) Conflict Driven Clause Learning short CDCL, (ii) Chronological backtracking to avoid blocking clauses and (iii) Implicant shrinking to minimize models.

Besides adding clauses iteratively, IPASIR-UP also allows to interfere with decision making, backtracking and found models. This raises the question whether it is possible to create a model enumerator through IPASIR-UP without blocking clauses by using all state-of-the-art techniques from TabularAllSat.

This thesis introduces CaDiCaL, a model enumerator which is almost entirely implemented through the IPASIR-UP interface and does not need any blocking clauses. "Almost" because there are two minor additions that need to be added to IPASIR-UP: One is already added to the newest version of CaDiCaL's IPASIR-UP interface and the other is fixing a bug introduced by the former. As IPASIR-UP is an interface for CaDiCaL and CaDiCaL being a CDCL-based solver, technique (i) from TabularAllSat is naturally covered. Also technique (ii) chronological backtracking is fully possible through the IPASIR-UP interface and thus also used in CaDiCaL. Using (iii) the implicant shrinking algorithm proposed by TabularAllSat requires access to the watch-lists of literals. This is not supported by the IPASIR-UP interface. However, it is still implemented in CaDiCaL and section ... discusses which additional functionality is needed and how it could be added to the IPASIR-UP interface.

CaDiCaL supports a fixed variable order. However, per default CaDiCaL decides. It also can be used as a simple model counter, which makes it possible to count models for formulas with exponential many models while keeping memory consumption very little. Furthermore, as IPASIR-UP also introduces the possibility to propagate literals. It is also possible to backtrack through adding reason clauses if necessary.

2 IPASIR

The IPASIR interface standardizes incremental interaction with CDCL-based SAT solvers by defining a minimal set of functions through which the user can call the solver in bigger contexts while keeping the program Independent of the solver. A solver instance is created and subsequently manipulated only via functions such as `ipasir_add` to add literals to the current clause, terminated by 0. `ipasir_assume` to add temporary assumption literals for the next call to `ipasir_solve`, and `ipasir_solve` to start or resume the search. After `ipasir_solve` returns SAT or UNSAT, the user may query models with `ipasir_val` or inspect a failed core via `ipasir_failed`. The interface deliberately abstracts from internal solver mechanisms: There is no access

to decision levels, propagation events, learned clauses, or backtracking. Although learned clauses are preserved across calls in incremental mode, the user cannot intervene during the search itself. Consequently, model enumeration requires blocking clauses: After finding a model, add its negation as a new clause to the cnf (via `ipasir_add`) and restart the solver. If the solver returns UNSAT, all models are found. However, this approach is sub-optimal as each new blocking clause increases the cnf. This makes it infeasible for big CNFs with many models. Model enumeration without blocking clauses through IPASIR is impossible since the solver terminates immediately upon finding a satisfying assignment.

3 IPASIR-UP

To overcome the limitations from IPASIR, Katalin Fazekas et al. introduced IPASIR-UP in 2023 as an extension of IPASIR that introduces **user propagators** (UP). IPASIR-UP augments the classical interface with a bidirectional communication layer between the solver’s CDCL loop and user-defined callbacks. Instead of restricting interaction to clause addition and top-level solve calls, IPASIR-UP enables controlled intervention during propagation, decision making, backtracking, and model detection.

- Model validation

```
bool cb_check_found_model(const tmodel &model) override {
    return true;
}
```

This function is called whenever the solver derives a total satisfying assignment. If the user propagator returns `true`, the solver reports SAT as usual. If it returns `false`, the model is rejected and the solver continues the search without restarting. Crucially, the internal trail, learned clauses, and heuristic state remain intact.

- External clause injection

```
bool cb_has_external_clause(bool &is_forgettable) override {
    return false;
}

int cb_add_external_clause_lit() override { return 0; }
```

These functions allow the user propagator to provide clauses to the solver right after `cb_check_found_model` was called. When `cb_has_external_clause` returns `true`, the solver repeatedly calls `cb_add_external_clause_lit` to obtain the literals of the clause (terminated by 0).

- Notifications

```
void notify_assignment(const ivec &list) override {}
void notify_new_decision_level() override {}
void notify_backtrack(size_t new_level) override {}
```

These notification callbacks inform the user propagator about changes to the assignment trail. `notify_assignment` communicates newly assigned literals, `notify_new_decision_level` signals that a decision was made and thus the decision level increased, and `notify_backtrack` reports to which level the solver backtracked.

- Decision making

```
int cb_decide() override { return 0; }
```

This callback is invoked before the solver makes the next decision. If it returns a non-zero literal, the solver uses it as the next decision. If 0 is returned, the solver makes its own decision, based on its internal decision heuristic.

- User propagation

```
int cb_propagate() override { return 0; }
int cb_add_reason_clause_lit(int propagated_lit) override {
    return 0;
}
```

`cb_propagate` enables the user propagator to return implied literals under the current partial assignment. If a conflict is detected in which the propagated literal was involved, the solver asks for a reason clause through repeatedly calling `cb_add_reason_clause_lit` (terminated by 0). This enables steering through the decision space while adding the reason clauses only if really necessary.

A user must define a User Propagator defining all the functions above.

```
void connect_external_propagator (ExternalPropagator* propagator);
void disconnect_external_propagator ();
void add_observed_var (int var);
void remove_observed_var (int var);
```

Then User Propagator can simply be connected and disconnected to the solver by calling `connect_external_propagator` and `disconnect_external_propagator`.

The notification callback functions are only called for variables marked as `observed`. The user can mark and unmark variables through the functions `add_observed_var` and `remove_observed_var`.

4 CaDiCAll

CaDiCAll is a fully functional model enumerator using CaDiCaL almost exclusively through the IPASIR-UP interface. It is implemented by using all three state-of-the-art mechanisms defined by Spallitta et al[3]: (i) Conflict Driven Clause Learning (Section 4.1), (ii) Chronological Backtracking (Section 4.2) and (iii) Implicant Shrinking (Section 4.3).

4.1 Conflict Driven Clause Learning

Conflict-Driven Clause Learning (CDCL) improves the classical DPLL algorithm by learning from conflicts. When the solver runs into a conflict, it does not simply flip the last decision but analyses the conflict to determine the assignments that caused the contradiction.

The analysis starts with the conflicting clause (the clause that is falsified by the current assignment). The solver inspects which literals in that conflict clause were forced by another clause and resolves them with each other. Let the conflict clause be $(A \vee \neg x_1)$ and x_1 be forced by $(B \vee x_1)$. Resolving these two clauses on x_1 yields the reason clause $(A \vee B)$. This process is repeated until a learned clause of the form $(A \vee B \vee \dots)$ is obtained that contains exactly one literal from the current decision level, which is then added to the CNF. Such a clause enables non-chronological backtracking: the solver backtracks to the second highest decision level occurring in the clause. At that point, all other literals in the clause remain falsified while the literal from the previous decision level becomes the only unassigned one, making the clause unit and forcing a new assignment through unit propagation. This prevents the solver from encountering the same conflict again.

By learning new clauses and pruning conflicting regions of the search space, CDCL significantly improves the efficiency of SAT solving. For this reason, it forms the basis of most modern SAT solvers.

As CaDiCaL is a CDCL-based solver, CDCL is implicitly provided to CaDiCAll when interfacing with CaDiCaL via IPASIR-UP.

4.2 Chronological Backtracking

Model enumeration counts every model exactly once. To do so, we must ensure, that paths which lead to models are excluded from future exploration. In common model enumerators this is done by blocking clauses, which permit the solver to run

into the same model. Without blocking clauses we must ensure manually that every path is explored exactly once.

Modern SAT solvers based on Conflict-Driven Clause Learning (CDCL) employ non-chronological backtracking (NCB). During conflict analysis, the solver may determine that some earlier decisions did not contribute to the conflict and can therefore backtrack across multiple decision levels at once. By skipping parts of the search tree that cannot lead to a solution, this approach significantly improves efficiency, but it also changes the order in which decisions are revisited and effectively reshapes the search tree during the search. NCB can lead to redundant work, which in normal Sat is not a problem, as the solver stops after finding one model and thus rediscovering the same model is not a problem.

In contrast, chronological backtracking (CB) always undoes only the most recent decision and revisits decisions strictly in reverse order, assigning the opposite value or continuing propagation. As a result, the structure of the search tree remains fixed. This difference becomes particularly relevant when enumerating all models, where it is essential to ensure that each model is reported only once. Because NCB can jump back to arbitrary earlier decision levels and dynamically alter the exploration order, the solver must explicitly prevent rediscovering the same assignment after a model is found. In practice, this requires excluding the complete decision path that led to the model so that the solver cannot revisit the exact same assignment again.

(TODO: why NCB and not CB)

In CadiCaL two reasons may trigger backtracking: (i) A model was found or (ii) a conflict occurs. Backtracking after a model was found is handled by the user propagator. Backtracking after a conflict is handled by the solver.

4.2.1 Backtracking after a model is found

Backtracking after a model is found was not supported by the IPASIR-UP interface. However, as discussed in Section X, allowing to backtrack after a model is found does not introduce any bad behavior and thus was added to the newest version of CaDiCaL's IPASIR-UP interface by Katalin Fazekas.

The idea is simple: After a model is found, determine the highest decision level `d1` with a decision that can be flipped and backtrack to the decision level `d1 - 1`. If this level is negative, the solver can terminate as there is no path left to explore.

- Determine the decision level to backtrack to

The order in which decisions are made as well as their phase can be chosen freely as described in Section X. In order to keep track which decision can be flipped and which have already been, for each decision level a counter, as well as the decision variable is saved. A decision count of one represents a decision that has been decided once, while a decision count of two represents

that the decision has already been flipped. Thus, the highest decision level with a decision count of one is the highest decision level with a decision that can be flipped.

There are two different options to continue the search from here: (i) Propagate the negation of the decision at level `d1` via `cb_propagate`. (ii) Decide the negation of the decision at level `d1` via `cb_decide`. CaDiCAll implements both, but focuses on the latter as it does not require adding reason clauses for the propagated literal.

- (i) Continue the search with propagation

Continuing the search through propagating the negation of the decision via `cb_propagate` seems elegant. When backtracking both, the solver and the user propagator can simply go back to the last decision which a decision count of one and propagate its negation. If the solver backtracks again, it simply backtracks over the propagated literal, as it is not a decision anymore. If the propagated literal is inspected while conflict analysis, the solver asks for a reason clause via `cb_add_reason_clause_lit`. The reason clause for this literal is the negation of all decisions made on a lower level. Thus, reason clauses must not be stored but can be computed on the fly only if necessary. This bears two advantages: (i) The size of the reason clause is smaller than blocking clauses as it only contains decisions on a lower level. (ii) Reason clauses are only added if necessary, thus avoiding adding unnecessary clauses.

However, CDCL learns from conflicts. If a conflict occurs the solver backtracks automatically and derives a reason clause, depending on that, the solver may propagate other literals and not the negated decision. Let the old decision be `d`. There are three cases to consider: (i) The solver propagates the old decision `d` again. In this case the solver starts to explore the same path again. This can be detected directly in `notify_assignment`. However, the first it can be handled are either by `cb_decide` or `cb_check_found_model` as these are the only two functions that can force the solver to backtrack. The function `notify_assignment` marks the current path as a duplicate and as soon as either `cb_decide` or `cb_check_found_model` is called, it forces the solver to backtrack further. This is sound as everything that led to the propagation of `d` was propagated itself. Thus, the propagation of `d` is a direct consequence of the conflict and implies that $\neg d$ always leads to a conflict. (ii) The solver runs into a conflict while propagating. In this case, a conflict induced backtracking occurs. (iii) The solver only propagates literals which are different from `d` or non at all. In that case, `cb_decide` will be called again. In `cb_decide` the user propagator detects that there is a decision that got backtracked but that negation has yet not been assigned. Thus, the user propagator decides $\neg d$. However, if the solver backtracks now again, it cannot backtrack simply to the

last decision but to the last decision that was only decided once. Otherwise the solver would rediscover the path with `d` again.

While this approach does optimize blocking clauses, it still uses them. Thus, CaDiCAll focuses on the second (ii) approach, which does not require adding any clauses at all. Running CaDiCAll with the first approach is still possible by running it with the option `--reason` and is also included in the benchmarks in Section X.

- (ii) Continue the search with decisions

Continuing the search through deciding the negation of the decision via `cb_decide` is more straightforward. After backtracking to the last decision that can be flipped, the user propagator simply decides the negation of that decision. This does not require adding any clauses at all. However, before `cb_decide` is called, the solver propagates. Let `d` be the old decision. This can lead to four different outcomes: (i) The solver propagates the old decision `d`. Thus, the solver starts to rediscover the old path. This is again detected directly in `notify_assignment` but can be handled the earliest when either `cb_decide` or `cb_check_found_model` is called. There, the solver is forced to backtrack further. (ii) The solver runs into a conflict while propagating. In this case, a conflict induced backtracking occurs. (iii) The solver only propagates variables that are different to decision `d` or none at all. In that case the user propagator can continue normally and return the flipped decision in `cd_decide`. (iv) The solver propagates $\neg d$ itself. Then the user propagator does not need to decide it manually and can let the solver chose the next decision itself by returning 0 in `cb_decide`.

4.2.2 Conflict induced backtracking

When the solver runs into a conflict and backtracks, the user propagator cannot interfere and is simply informed about it via `notify_backtrack`.

How the solver backtracks is an option that can be set through IPASIR-UP. In CaDiCAll it is set to chronological backtracking. Thus, backtracking after a conflict preserves the chronological order and does not alter the search space.

- Determine the decision level to backtrack to

As backtracking is set to chronological, the solver always backtracks to the last decision.

Always backtracking to the last decision is problematic. No matter how we continue the search, with propagations or with decisions, in both the last decision is not always the one we want to flip: (i) When using propagations, the solver may not propagate the negation of the last decision and thus it is forced via `cb_decide`.

In that case the decision count is two, indicating that this decision has already been flipped. (ii) When using decisions to flip the last decision, the decision counter is two as well. In both cases we want to backtrack further, to the last decision with a decision count of one. However, as the solver backtracks on its own, it backtracks to the last decision, no matter what decision count it has. If the decision count was two, everything the solver does from this moment on must be undone again. This case is detected directly in `notify_backtrack` and handled by the user propagator in either `cb_decide` or `notify_backtrack`. Unfortunately, this can lead to an overhead as the solver may spend some time propagating.

When the decision count is one, the same four cases as in the backtracking after a model was found can occur: (i) The solver propagates the old decision `d`. Thus, the solver starts to rediscover the old path. This is again detected directly in `notify_assignment` but can be handled the earliest when either `cb_decide` or `cb_check_found_model` is called. There, the solver is forced to backtrack further. (ii) The solver runs into a conflict while propagating. In this case, a conflict induced backtracking occurs. (iii) The solver only propagates variables that are different to decision `d` or none at all. In that case the user propagator can continue normally and return the flipped decision in `cd_decide`. (iv) The solver propagates $\neg d$ itself. Then the user propagator does not need to decide it manually and can let the solver chose the next decision itself by returning 0 in `cb_decide`.

4.3 Deciding

The order in which decisions are made can heavily impact the performance of sat solvers. The right decision can lead the solver into early conflicts, pruning big parts of the search space completely or skipping long nested parts. While in model enumeration the later benefit is lost as all models must be found, it still profits from early conflicts.

In order to search systematically through the whole search tree, a fixed variable order is the easiest. On each `cb_decide` call, decide the next undecided variable positively. If the solver needs to backtrack, simply find the highest decision level with a positive decision. However, this constraint can be weakened while still preserving a systematically order.

First, the phase of a decided variable can be chosen freely, while keeping backtracking still simple enough. For each decision level a decision count is saved. Instead of backtracking to the highest decision level with a positive decision, simply backtrack to the highest decision level with a decision count of 1. Meaning it has been decided exactly once.

Second, the order in which variables are decided can also be chosen complete free, as long as after a backtracking the decision is just flipped. This enables the solver to profit from dynamic decision heuristics such as Variable State Independent Decaying

Sum (VISIDS) which is known to improve solver performance significantly and thus is a commonly used decision heuristic in modern sat solvers. The performance difference between fixed and dynamic decision orders can be seen in

The decision count is updated if a decision is made or flipped and reset if the decision is propagated or backtracked over. If the solver asks for a decision and the decision count on the current level is one. The decision must be flipped and thus `cb_decide` returns the negation of the old decision. Otherwise, `cb_decide` returns 0 and thus, lets the solver decide itself.

4.4 Implicant Shrinking

Models (also called implicants) found by the solver are total. Models are total if they contain every variable. For a CNF $(x_1 \vee x_2) \wedge (x_1 \vee \neg x_2)$ a model enumerator returns two total models: $x_1 \wedge x_2$ and $x_1 \wedge \neg x_2$. As x_1 occurs in both models positively, the only difference lays in x_2 which occurs once positively and once negatively. In other words, if x_1 is positiv, it does not matter what the value of x_2 is, it is always a model for the CNF. Thus, these models can be merged into one model by dropping x_2 , resulting in one partial model x_1 . This reduces both, the amount of models in total and the size of models. It also may fasten the solving itself. Normally, after finding the first model $x_1 \wedge x_2$ a model enumerator backtracks to the last decision that can be flipped. Assuming the decisions where x_1 at decision level 1 and x_2 at decision level 2, a model enumerator would backtrack to decision level one and then continue with $\neg x_2$. However, if the model enumerator shrinks the model $x_1 \wedge x_2$ to x_1 the last decision that can be flipped is x_1 . Thus, the model enumerator backtracks to decision level 0 and can continue with $\neg x_1$, skipping the branch with $\neg x_2$ completely. As the branch containing x_2 could be arbitrary large, this may reduce the search space significantly.

Finding out which literals can be dropped is non trivial. Implicant shrinking algorithms from non chronological backtracking based model enumerators do not guarantee mutual exclusiveness between different assignments[3]. Spallitta et al. [3] introduces their own shrinking algorithm that is (i) compatible with chronological backtracking and (ii) tries to cut the highest amount of literals while still ensuring mutual exclusivity. This algorithm starts with the last literal on the Trail and tries to drop as many literals as possible. A literal ℓ can be dropped if ℓ is a decision and all clauses in which ℓ occurs contain another literal ℓ' that satisfies the clause.

In CaDiCaL the 2-watched-literal scheme is used. This allows the solver not to constantly check all clauses if the assignment of a literal changes. Instead every clause of the CNF is only watched by two literals, let them be ℓ_1 and ℓ_2 . This is enough to track the status of the clause if the following invariant holds: At least one literal is not falsified. In order to keep this invariant alive, if one watched literal ℓ_1 becomes falsified the solver tries to find a replacement. If there is another literal

ℓ' in that clause that is neither false nor ℓ_2 , it replaces ℓ_1 as the watching literal. If there exists no such literal ℓ' the solver knows that ℓ_2 is the only literal left that could satisfy the clause and forces ℓ_2 . If both literals ℓ_1 and ℓ_2 are falsified, the clause is falsified, yielding UNSAT.

If we want to drop a literal ℓ , it suffices to check all clauses that are watched by ℓ instead of all clauses in which ℓ occurs. If a model is found, we know at least one of the watching literals for every clause is satisfied. Thus, dropping a literal that is not watched will never falsify a clause. However, if we drop a literal ℓ that is currently watching a clause c , we know there exists another literal $\ell' \in c$ that satisfies c . This literal ℓ' must now become the new watching literal for c . Otherwise, when trying to drop ℓ' , c would not be checked.

Algorithm 1 IMPLICANT-SHRINKING(T)

```

1:  $b \leftarrow 0$ 
2:  $T' \leftarrow T$ 
3: while  $T' \neq \epsilon$  do
4:    $\ell \leftarrow T'.\text{pop}()$ 
5:   if  $\rho(\ell) \neq \text{DECISION}$  then
6:      $b \leftarrow \max(b, \delta(\ell))$ 
7:   else if  $\delta(\ell) > b$  then
8:      $b \leftarrow \text{CHECK-LITERAL}(\ell, b, T')$ 
9:   else if  $\delta(\ell) = 0$  or  $(\delta(\ell) = b \text{ and } \rho(\ell) = \text{DECISION})$  then
10:    break
11:   end if
12: end while
13: return  $b$ 

```

Going through all literals in all clauses watched by ℓ may introduce a bottleneck. Thus Spallitta et al [3] introduced a faster but more conservative algorithm exploiting the 2-watched-literal scheme. Conservative meaning it may not drop all literals that could be dropped.

The 2-watched-literal scheme is used in the faster approach proposed from Spallitta et al. [3] as shown in algorithm 2.

Algorithm 2 CHECK-LITERAL(ℓ, b, T') (faster but more conservative)

```

1: for all  $c \in \omega(\ell)$  do
2:    $\ell' \leftarrow$  the other watched literal in  $c$ 
3:   if  $\ell' \in T'$  then
4:     Continue
5:   else
6:      $b \leftarrow \max(b, \delta(\ell))$ 
7:   end if
8: end for
9: return  $b$ 

```

Instead of checking all other literals in the watched clause, simply check the other literal ℓ' that watches that clause. If it satisfies the clause as well, the literal ℓ can

be dropped. As ℓ' already watches the clause, nothing must be updated. This is more conservative as it may not drop literals that could be dropped. If ℓ' is falsified but another literal in that clause is satisfied, ℓ still could be dropped but is not. However, as this conservative approach does not need to traverse the clause and check each literal, it is significantly faster.

As IPASIR-UP does not allow any access to watches, this is implemented by using a backdoor to CaDiCaL directly. Section XX discussed what is needed and how it could be added to the IPASIR-UP interface.

Adapting this implicant shrinking algorithm to CaDiCaL using IPASIR-UP may introduce a fault. As stated in Section 5.1 and 5.2 there are two different ways to flip a decision. If the decision is flipped by deciding its negation, algorithm 1 must be adapted.

Consider the CNF $(x_1 \vee x_2 \vee x_3)$. Let x_2 and x_3 be the watched literals at the start and the decision order be $x_1 \rightarrow x_2 \rightarrow x_3$. The first model to be found will be $x_1 \wedge x_2 \wedge x_3$. Implicant shrinking will first remove x_3 as the other watched literal (x_2) still satisfies the clause. Then it will try to remove x_2 . However, as the other watched literal is x_3 which is no longer on the trail, it cannot continue. Thus, discovering the model $x_1 \wedge x_2$. Then the solver backtracks and flips x_2 by deciding its negation $\neg x_2$. Remind that $\neg x_2$ is still a decision. As x_2 is one of the watched literals but is now falsified, x_1 becomes the new watched literal, as it is the only literal in the clause that is neither false nor already watched. Then the solver decides x_3 resulting in the model $x_1 \wedge \neg x_2 \wedge x_3$. Now implicant shrinking starts again. First removing x_3 as the other watched literal (x_1) still satisfies the clause. Next it tries to remove $\neg x_2$ as it is the next highest decision. As $\neg x_2$ does not occur in any clauses, it cannot be a watched literal. Thus removing it does not falsify any clause. Then, it tries to remove x_1 . However, as the other watched literal is x_3 which was already removed, x_1 can not be removed. Hence, yielding the model x_1 .

The implicant shrinking algorithm yielded both models $x_1 \wedge x_2$ and x_1 . These models are not distinct, as setting x_2 in the later generates the former.

This can be fixed by allowing implicant shrinking only to remove literals with a decision count of 1 (line 5 in algorithm 3). Consider a variable with decision count 2 to be checked by implicant shrinking. There are two possibilities: (i) The variable could not be removed when it's decision count was 1. Removing it now could trigger exactly a case like the example above. (ii) The variable was removed when it's decision count was 1. In that case the model enumerator would have backtracked over the variable and thus resetting the decision count, which makes this case impossible to reach.

Flipping decisions by propagation their negations (Section 5.2) is not affected by this fix, as the highest decision count is always one.

Algorithm 3 IMPLICANT-SHRINKING(T)

```
1:  $b \leftarrow 0$ 
2:  $T' \leftarrow T$ 
3: while  $T' \neq \epsilon$  do
4:    $\ell \leftarrow T'.\text{pop}()$ 
5:   if  $\rho(\ell) \neq \text{DECISION}$  and  $\text{decision\_count}(\ell) = 1$  then
6:      $b \leftarrow \max(b, \delta(\ell))$ 
7:   else if  $\delta(\ell) > b$  then
8:      $b \leftarrow \text{CHECK-LITERAL}(\ell, b, T')$ 
9:   else if  $\delta(\ell) = 0$  or  $(\delta(\ell) = b \text{ and } \rho(\ell) = \text{DECISION})$  then
10:    break
11:   end if
12: end while
13: return  $b$ 
```

5 Extending IPASIR-UP

The idea of CaDiCall is to use IPASIR-UP as the only interface to the solver. This means that all features needed for model enumeration must be implemented using IPASIR-UP. This has the advantage that CaDiCall can be easily adapted to other solvers supporting IPASIR-UP without any changes to the solver itself. However, IPASIR-UP does not provide all features needed for an efficient model enumerator. Thus, some extensions to IPASIR-UP are necessary.

We propose three kinds of extensions to IPASIR-UP: (i) Additional features required to make model enumeration using IPASIR-UP possible. (ii) Additional features to allow implicant shrinking and (iii) Additional features to avoid duplications and thus making CaDiCall faster using IPASIR-UP.

5.1 Making IPASIR-UP the only interface needed

5.1.1 Backtracking after a model was found

In conventional SAT solving, backtracking only occurs when the solver encounters a conflict. To continue the search, the solver must undo at least one previous decision and explore a different branch of the search space.

However, a similar situation arises when enumerating all satisfying assignments (models). Once the solver finds a model, the current assignment satisfies all clauses and therefore no conflict is present to trigger backtracking. Nevertheless, the search must continue in order to determine whether additional models exist. Thus, the solver must be forced to leave the current branch of the search space and explore alternative assignments.

When a model is found, IPASIR-UP allows the user propagator to observe and reject the model by returning **false** in `cb_check_found_model`. It then calls `cb_has_external_clause`. If the user propagator returns **true**, the solver repeat-

edly calls `cb_add_external_clause_lit` allowing the user propagator to add the negation of the model as a blocking clause. This forces the solver to backtrack and explore other branches of the search space. However, as we do not want to add blocking clauses, CaDiCaL always returns `false` in `cb_check_found_model`. Which means, the solver is not forced to backtrack and would report the same model directly again. Resulting in an infinite loop where the solver is stuck.

In order to allow the user propagator to maintain additional constraints, IPASIR-UP provides the function `force_backtrack`. This function is allowed to be called inside `cb_decide`. It enables the user propagator to force the solver to backtrack to a specific decision level before making a new decision. Forcing the solver to backtrack inside `cb_decide` does not introduce bad behavior as at this point, the trail is fully consistent: All assignments up to the current decision level do not falsify the CNF, they are not in conflict with any other propagations and no propagations are pending.

When calling `cb_check_found_model` the trail is in an equally consistent state as in `cb_decide`: The trail fully satisfies the CNF and as it is a total assignment, there are no pending propagations. Thus, allowing `force_backtrack` in `cb_check_found_model` would not introduce any bad behavior.

To allow `force_backtrack` to be called from within `cb_check_found_model` simply the flag `forced_backt_allowed` must be set to `true` in the solver before calling `cb_check_found_model`.

```
forced_backt_allowed = true;
bool is_consistent = external->propagator->cb_check_found_model (etrail);
forced_backt_allowed = false;
```

This feature was added to the newest version of CaDiCaL's IPASIR-UP interface by Katalin Fazekas. Thus, allowing CaDiCaL to backtrack to the last decision after a model is found and continuing the search from there without adding blocking clauses.

5.1.2 Maintaining chronological backtracking

When the solver is about to make the next decision, the solver internally calls `ask_decision`, which then calls `cb_decide`. The user propagator is allowed to call `force_backtrack` inside `cb_decide`. However, backtracking might make the decision returned by the user propagator invalid, as it might be already assigned due to propagations after backtracking. However, instead of asking the user propagator again, the solver decides then on its own. This imposes a problem as CaDiCaL does not allow all decision to be made by the solver, some decisions are already fixed in order to force the solver to explore the search space chronologically. This can be fixed by calling `ask_decision` recursively instead of letting the solver decide. This

way, the user propagator can make a new decision after backtracking and thus keep the exploration of the search space chronological.

An important note is that, if the user propagator backtracks below the pseudo decision level of the solver. The solver goes back to the CDCL loop instead of making a decision. In that case the solver ignores the decision made by the user propagator, continues with the CDCL loop and calls `ask_decision` again when it is about to make the next decision. However, the CDCL loop may propagate literals that lead to paths that have already been explored. Thus, the user propagator must check whether the work done by the CDCL loop has led to such a path and if so, backtrack further. If not, the user propagator can either return the original decision or make a new one if the CDCL loop already propagated it.

```
if ((size_t) level < assumptions.size ()
    ((size_t) level == assumptions.size () && constraint.size ())) {
    return 0;
} else {
    return ask_decision();
};
```

5.1.3 Out of order propagations

In order to keep blocking literals correct, CaDiCaL propagates units at the root level to derive all implied units. However, this allows the solver to backtrack multiple levels and thus, skipping branches that potentially lead to models.

```
bool Internal::propagate_out_of_order_units () {
    return true;
};
```

5.2 Implicant Shrinking

The implicant shrinking algorithm described in section X needs access to the all watch-lists of a given literal. However, this is not supported via the IPASIR-UP interface. The implicant shrinking algorithm uses them read only. Thus, exposing a read only reference to them is unproblematic and could be integrated into IPASIR-UP without any changes to CaDiCaL itself. Here it is important to note that literals can be named differently external and internal. However, there already exist functions to translate from internal to external and vice versa, and thus the internal representation can be completely hidden through the IPASIR-UP interface.

A function implementing this feature would look like the following. It takes an external variable as input and returns all other literals that are currently watching

the same clauses.

```
std::vector<int> other_watching_literals(int external_variable) {
    int internal_variable = internal->external->e2i[external_variable];
    int internal_literal = internal->external->vals[internal_variable] ?
    ↪ internal_variable : -internal_variable;

    std::vector<int> other_watching_literals;
    for (CaDiCaL::Watch watch : internal->watches(internal_literal)) {
        int i_wl1 = watch.clause->literals[0];
        int i_wl2 = watch.clause->literals[1];
        int i_other_lit = internal_literal == i_wl1 ? i_wl2 : i_wl1;
        int e_other_lit = internal->externalize(i_other_lit);
        other_watching_literals.push_back(e_other_lit);
    }

    return other_watching_literals;
}
```

The user propagator can then check whether all other watching literals are satisfied by the remaining trail. If so, the literal can be dropped without falsifying any clause. If not, the literal cannot be dropped as it is still needed to satisfy at least one clause.

If the sole purpose of this function is to be used by the implicant shrinking algorithm implemented from CaDiCaL, it could be optimized by checking if the other watching literal satisfies the clause instead of appending it to the list of other watching literals. This way, the function can return a boolean value indicating whether the literal can be dropped or not, without the need to return all other watching literals. This also allows early breaks, as the function can return false as soon as it finds a clause that is not satisfied by the other watching literal. However, the function would also need either access to the internal stack and position of literals, as the user propagator needs to check if the other watching literal is satisfied by the remaining trail, or it would need to be passed the current trail as an argument.

With this minor addition to the IPASIR-UP interface, the implicant shrinking algorithm can be implemented entirely through IPASIR-UP.

5.3 Optimization

This thesis aimed to show if and how a model enumerator using IPASIR-UP is possible. How CaDiCaL and the IPASIR-UP interface can be optimized to reduce CaDiCaL's runtime was explicitly looked at. However, there are some obvious optimizations worth mentioning.

5.3.1 Interacting with the solver

When the solver calls the functions `notify_assignments` and `cb_check_found_models` it creates a vector containing assigned literals. When we timed CaDiCall, in CNFs with many models, CaDiCall spend around 40 percent of its total time with vector operations. When passing these vectors from the solver to the user propagator, the solver always allocates memory for the vectors and frees it afterwards again. By making these vectors persistent, the total running time of CaDiCall on CNFs with many models was cut in half.

5.3.2 Duplicating Datastructures

In order to check where to backtrack to, if the solver recovers paths that have already been explored, or to check which literals can be dropped during implicant shrinking, CaDiCall needs to maintain the values of variables, their decision levels and whether they are decisions or propagations. IPASIR-UP provides through the notification callback functions `notify_assignment`, `notify_new_decision_level` and `notify_backtrack` enough information to maintain these values external from the solver itself. However, this means that every time a variable is assigned, it is not only stored internally by CaDiCaL but also externally by CaDiCall. This does not only double the expenses but introduces an even bigger overhead as the information must also be passed from the solver to the user propagator through the callback functions. Thus, allowing read only access to CaDiCaLs variable assignment and stack would make most external representation redundant.

6 Results

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7 Conclusion

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References

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Freiburg im Breisgau, den 01.01.2016.